

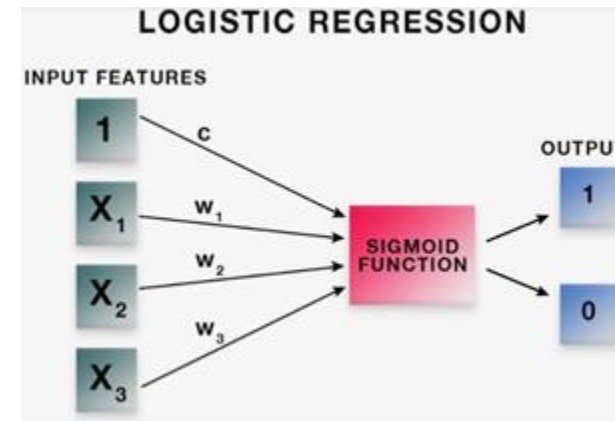
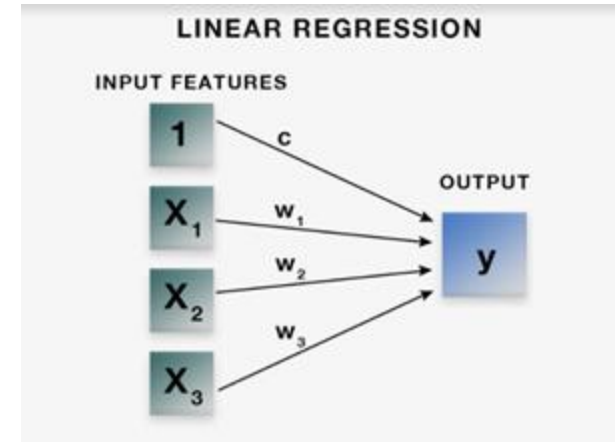
Logistic Regression

Is it Classification or Regression

Logistic Regression is a Supervised and Classification Algorithm

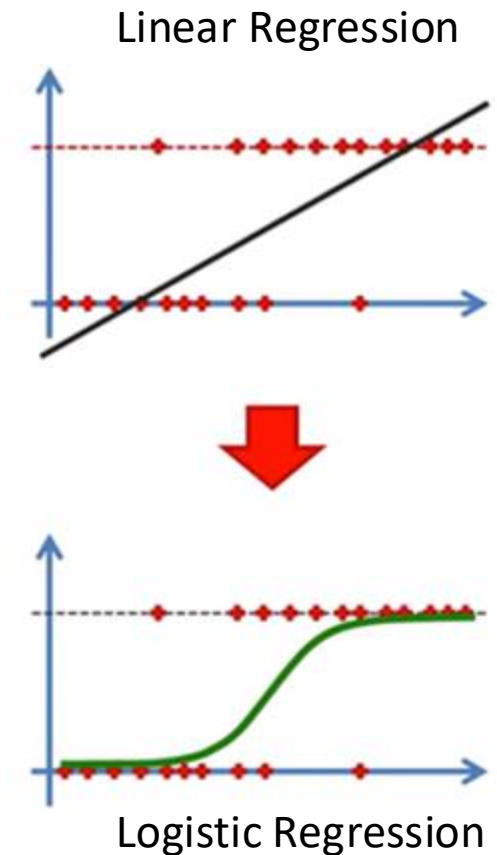
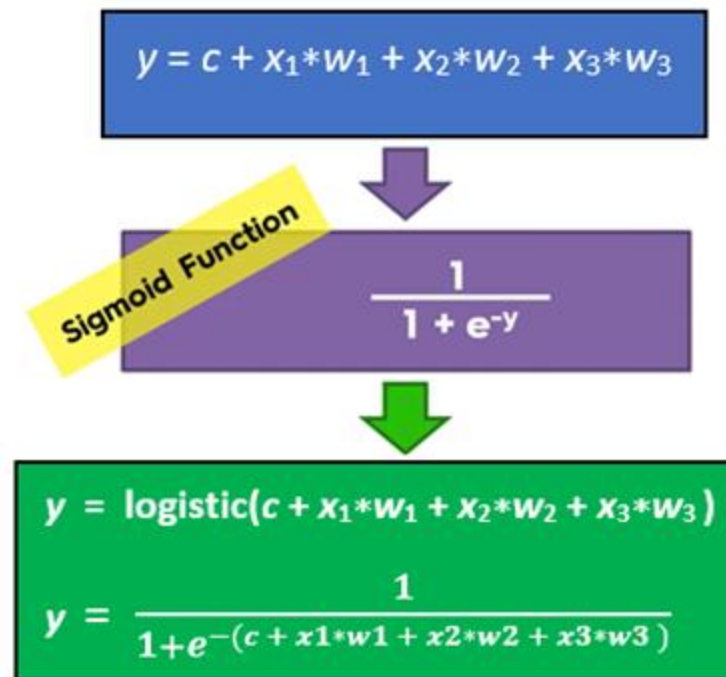
Why Logistic Regression called as “Regression”

- In linear regression, a real-valued output y is predicted based on a weighted sum of input variables.
- Whereas in Logistic regression, transforms its output using the logistic sigmoid function to return a probability value of 1 or 0.



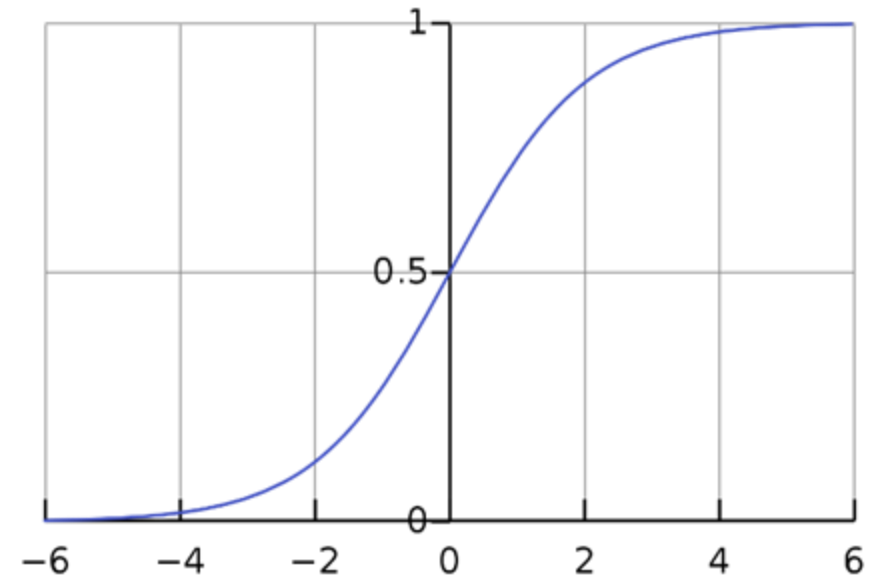
Transform the model

- Logistic Regression is a type of Generalized Linear Models.
- Output of the linear regression is passed through a sigmoid function that can map any real value between 0 and 1.



Introduction

- Logistic Regression computes the probability which lies between the interval 0 and 1 (inclusive of both).
- For example, if the computed probability comes out to be greater than 0.5, then the data belonged to class A and otherwise, for less than 0.5, the data belonged to class B.



Know about Logistic Regression

- Predicts the categorical dependent variable using a given set of independent variables.
- Predictive modeling.
- Uses a more complex cost function called as Sigmoid function or logistic function.
- Maps predicted values to probabilities which lie between 0 and 1.
- The logistic regression out is the probability of base class in a binary classification problem.

Understanding the difference

➤ Linear Regression

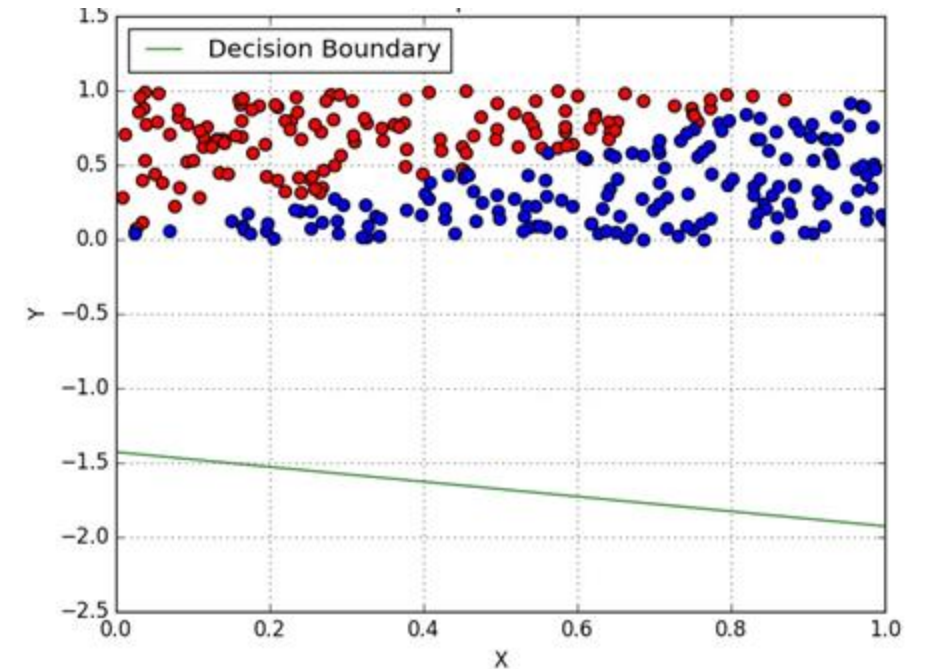
- Purpose to find the best fitted line
- Solves Regression Problems.
- Dependent variables are continuous in nature
- Independent variable can be correlated with each other
- Linear function represents value greater than 1 or less than 0.
- Loss function in linear regression is the mean squared error
- Output will be value of the variable.

➤ Logistic Regression

- Fits the line values to an "S" shaped logistic function.
- Solves Classification Problems.
- Dependent variable is categorical or binary in nature
- Independent variable must not be correlated with each other.
- Limits the Logistic function or Sigmoid function between 0 and 1.
- Loss function for logistic regression is Log Loss
- Output will be probability of occurrence of events.

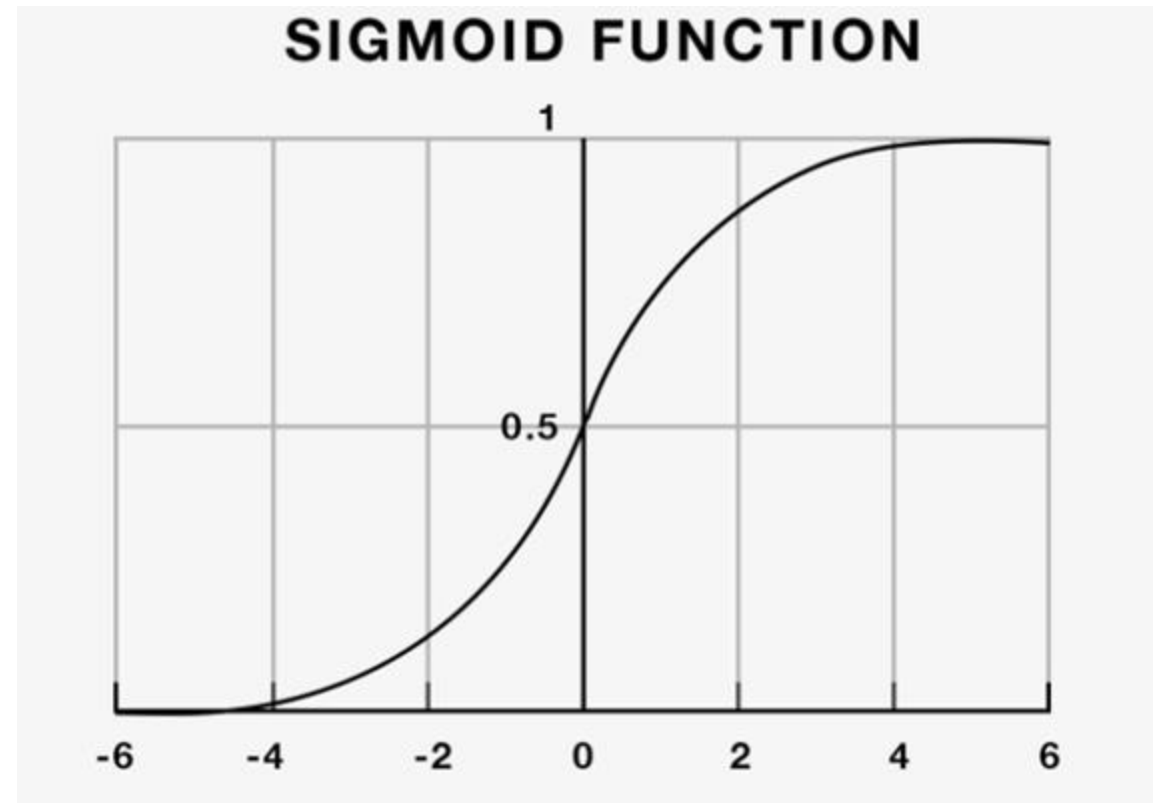
Decision Boundary

- In this fig, A boundary line created by the classifier (Logistic Regression) signifies the decision regions.



Sigmoid Function

- As you see in the graph, it is an S-shaped curve that gets closer to 1 as the value of input variable increases above 0 and gets closer to 0 as the input variable decreases below 0. The output of the sigmoid function is 0.5 when the input variable is 0.
- If the output is more than 0.5, we can classify the outcome as 1 (or positive) and if it is less than 0.5, we can classify it as 0 (or negative).



Hypothesis of Logistic Regression

- In Linear Regression, hypothesis represented in the form of

$$h_{\theta}(x) = \theta^T x$$

- Passing through the sigmoid function

$$h_{\theta}(x) = g(\theta^T x); \text{ where } g(z) = \frac{1}{1+e^{-z}}$$

$$h_{\theta}(x) = \frac{1}{1+e^{-\theta^T x}}$$

Probability with Sigmoid Function

- The transformation is defined as the logged odds:
 - Odds = probability presence of the characteristic (P) /probability of absence of characteristic (1-P)
 - $\ln(P/(1-P)) = c + x_1*w_1 + x_2*w_2 + x_3*w_3$
- Being positive class over being negative class could be expressed as below for a binary classification task.
$$\frac{P(y = 1)}{P(y = 0)}$$
$$\frac{P(y = 1)}{(1 - P(y = 1))}$$

Types of Logistic Regression

Based on those number of categories, Logistic regression can be divided into following types :

- **Binary or Binomial** - dependent variable have only two possible types either 1 and 0.
 - For example, these variables may represent success or failure, yes or no, win or loss etc.
- **Multinomial** - dependent variable can have 3 or more possible *unordered* types or the types having no quantitative significance.
 - For example, these variables may represent “Type A” or “Type B” or “Type C”.
- **Ordinal** - dependent variable can have 3 or more possible *ordered* types or the types having a quantitative significance.
 - For example, these variables may represent “poor” or “good”, “very good”, “Excellent” and each category can have the scores like 0,1,2,3.

Applications of Logistic Regression

- Predicting a probability of a person having a heart attack
- Predicting the probability of failure of a given process or product.
- Predict whether a bank loan can be granted to a particular customer or not.
- More....

Demo Experiment

- Demo Experiment on Logistic Regression
 - To find the logistic regression function $p(\mathbf{x})$ such that the predicted responses $p(\mathbf{x}_i)$ are as close as possible to the actual response.

Thanks!!
Questions?